

Clinical Risk Assessment Report: Patient 6

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Negative for Metastasis
- Prediction Confidence: Moderate Confidence
- Absolute Predicted Risk: 16.1% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 0.73x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as Low-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Tumor Sphericity (CT) (SHAPE_Sphericity(onlyFor3DROI)).CT), which reduced the patient's absolute risk by 2.1%. Biologically, this feature quantifies how closely the tumor resembles a perfect sphere; lower values indicate irregular, spiculated, or highly invasive margins.

2. Key Pathological & Imaging Drivers (Elevating Risk)

Large Zone Emphasis (CT) (GLZLM_LGZE.CT) Value: -0.32 [Bottom 19%] (+3.1%)

Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

Tumor Sphericity (PET) (SHAPE_Sphericity(onlyFor3DROI)).PET) Value: 0.59 [Avg (70th %ile)] (+1.1%)

Quantifies how closely the tumor resembles a perfect sphere; lower values indicate irregular, spiculated, or highly invasive margins.

Short Run Low Gray-Level Emphasis (CT) (GLRLM_SRLGE.CT) Value: -0.19 [Avg (26th %ile)] (+0.8%)

Indicates fragmented areas of low density, often associated with diffuse micro-necrosis or loose tissue structure.

3. Protective / Mitigating Factors (Reducing Risk)

Tumor Sphericity (CT) (SHAPE_Sphericity(onlyFor3DROI)).CT) Value: 0.68 [Top 24%] (-2.1%)

Quantifies how closely the tumor resembles a perfect sphere; lower values indicate irregular, spiculated, or highly invasive margins.

Patient Age (Age) Value: 68.0 [Avg (68th %ile)] (-1.4%)

Patient age influences immune surveillance, tissue microenvironment, and baseline metabolic rates.

75th Percentile Tumor Density (CONVENTIONAL_HUQ3) Value: 0.42 [Avg (55th %ile)] (-1.2%)

Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.

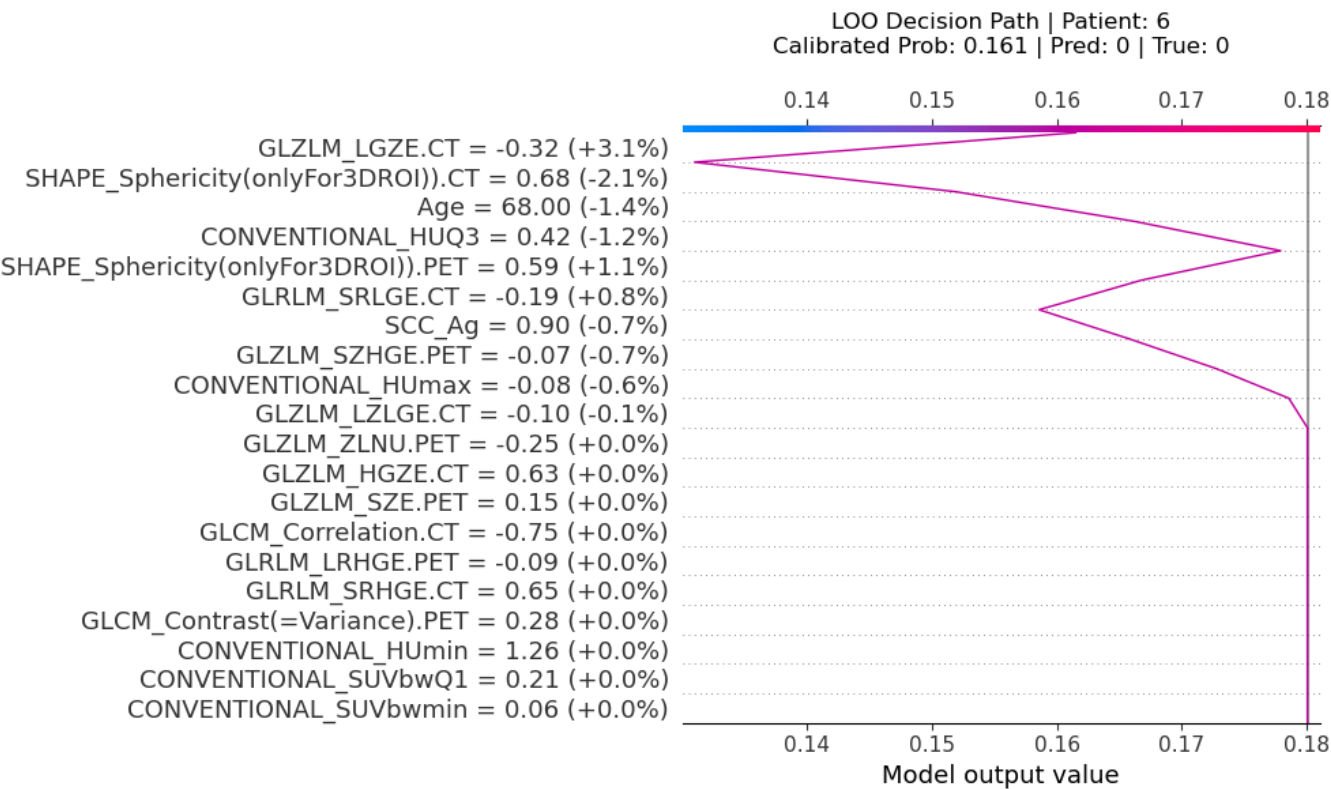
Squamous Cell Carcinoma Antigen (SCC_Ag) Value: 0.9 [Avg (74th %ile)] (-0.7%)

Serum levels reflect tumor proliferation, burden, and squamous cell differentiation.

Small Zone High Gray-Level Emphasis (PET) (GLZLM_SZHGE.PET) Value: -0.07 [Avg (73th %ile)] (-0.7%)

Indicates multiple small areas of intense density or metabolism, a marker of erratic and aggressive cellular proliferation.

Machine Learning Feature Attribution (SHAP Decision Path)



Sensitivity Analysis: Feature Tipping Points

The plots below demonstrate how variations in specific clinical or radiomic features affect the model's predicted risk. The red marker indicates the patient's current clinical state. Crossing the red dashed line changes the diagnosis.

